David Shumway

Software Engineer | Machine Learning | Backend & Data Systems | Remote

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Summary

Senior Software Engineer and Machine Learning Engineer with 10+ years of experience building production backend systems and applied ML pipelines. Master's graduate (CS, UIC) specializing in knowledge graphs, domain adaptation, and transfer learning for environmental and biomedical data. Track record of end-to-end project ownership, from software design and machine learning model development and deployment, to distributed data pipelines, ETL/ELT orchestration, and monitoring, across both research and client-facing environments.

Employment History

Graduate Research Assistant (ML / Knowledge Graphs), University of Illinois at Chicago - Chicago, IL *2021-2026*

- Lead developer on ONR-funded project (#00014-21-1-2286) applying ML to predict risk to Navy divers at coastal locations.
- Built and compared supervised, semi-supervised, and unsupervised models (LSTM, GRU, GNN, XGBoost, CNN, MLP, Random Forest) for time-series water quality prediction.
- Implemented domain adaptation and transfer learning methods (Feature Augmentation, Balanced Weighting, CORAL, subspace alignment) to generalize models to data-scarce coastal sites, published at IEEE BigData 2024, ES&T 2025.
- Designed and orchestrated ontology-driven ETL/ELT pipelines integrating environmental datasets (NOAA, EPA, WaterQualityData.us) for knowledge graph construction, published at ACM Web Conference 2023.
- Built OWL ontologies in Protégé for waterborne illness risk assessment; queried with SPARQL and RDF.
- Integrated large-scale public datasets (NOAA, EPA, NDBC, NREL, EN4) to improve coastal water quality ML models.

Graduate Research Assistant (Biomedical ML), University of Illinois at Chicago - Chicago, IL *2018-2021*

- Built MALDI-DB (NIH #R01GM125943), a community-driven public repository with ML workflows for MALDI-TOF spectra, deployed via Docker/OpenStack on MassOpenCloud (Python, Django, R, PostgreSQL, nginx, D3.js).
- Developed CNN-based models for bacteria identification in MALDI-TOF mass spectrometry data.
- Co-authored grant proposal for EarthCube semantic data infrastructure connecting distributed scientific datasets.

Teaching Assistant, University of Illinois at Chicago - Chicago, IL *2018-2026*

- Served as Instructor of Record for CS586: Data and Web Semantics, taking over mid-semester covering ontology engineering, RDF, OWL, and semantic web.
- Led discussions and mentored students across 11 semesters in courses including Big Data Mining, Software Design, Languages & Automata, and Mobile Development (groups of 5–250).
- Mentored 4 semesters of NSF-REU undergraduate researchers.

Senior Software Engineer, Scalesology - Remote *9/2024-2/2026*

- Led backend development for a production CRM platform (insurance client) including system design, PostgreSQL RDS on AWS, scalable API layer, and client-facing dashboards, and mitigated delivery risks related to system-level design gaps, while advocating for upfront architecture to address underestimation and scope creep.
- Migrated legacy SQL Server procedures to Python/Pandas services backed by Snowflake data warehousing workflows.
- Built interactive analytics dashboards and ETL/ELT data pipelines supporting client analytics workflows across energy, grocery, finance, and education sectors (ECharts, React, PowerBI, Tableau, D3.js).
- Implemented logging and observability for backend services supporting production CRM workflows on AWS.
- Delivered features in iterative releases with minimal supervision in a client-facing consulting environment.

- Coordinated with cross-functional stakeholders to translate requirements into scalable implementations.
- Leveraged LLMs such as GitHub Copilot daily for generating and reviewing GitHub PRs, creating API stubs, module system design, and producing skeleton code, with autocomplete in VS Code filling whole code blocks and careful review to ensure quality and integration.

Co-Owner & Lead Developer, Finger Lakes Business Systems - Remote *2010-2018*

- Collaborated with 750+ clients on Amazon MTurk, completing 3.3M+ microtasks across 1100+ unique projects with $185K revenue while maintaining a 99.98% task approval rating, continuing off-site with key clients.
- Engineered in-house web scraping, data management, workflow optimization, and distributed systems solutions, writing 375K+ lines of in-house code in JavaScript, PHP, HTML/CSS, SQL, and Python, with extensive use of Google Chrome API and browser extension programming.
- Created and maintained a simple open-source browser extension to help streamline the MTurk worker interface, published to the Chrome Store and Firefox Add-Ons, reaching up to 10K weekly users.
- Designed factory-based browser extension architecture (~300K LOC JS) with tab-pipelined execution, preview-mode admission control, and cross-worker disk-cached scraping, achieving effective average hourly wages 3-6x above published worker benchmarks, with a MySQL backend storing 100s of millions of records (scraper data, task payloads, completion logs).
- Designed 300+ reusable library methods and 110+ factory methods (JavaScript), enabling rapid deployment of new task types by assembling pre-built components rather than writing custom code from scratch
- Built 150+ PHP API endpoints for backend services, from file existence checks, disk-cached request utilities, server-side queueing with file locking, and SQL execution, to PDF-to-image conversion, and PDF-to-OCR, exposing numerous server-side capabilities to browser extensions

Summer Intern, École Polytechnique Fédérale de Lausanne, DSIL - Lausanne, Switzerland *2017*

- Enhanced Kamusi.org's Neo4j natural language database by analyzing and integrating new datasets.

Digital Archivist & Technical Specialist, Paul Brunton Philosophic Foundation - Burdett, NY *2010-2014*

- Curated 100K+ scanned documents and designed LAMP, JavaScript/HTML, Bash, and VBA solutions.
- Automated scanning process, resulting in 4-8x efficiency and saving 2-4 years of manual work.

Prior to 2010, held various roles including freelance web development (PHP/MySQL/JavaScript/HTML/CSS), audio transcription, food delivery, retail, and manual labor while self-teaching programming and building foundational technical skills.

Education

M.S. in Computer Science, University of Illinois at Chicago (August 2026) *2018-5/2026*

- Advisors: Isabel Cruz (in memoriam), Cornelia Caragea.
- Completed PhD qualifying exam in *Advances in Scientific Workflow Management System Provenance Storage and Querying* (2020). 1.5 oral, 2.0 written (1-5, 1=best).
- Completed PhD coursework requirement (2021). Transitioned to M.S. program Summer 2026 with coursework-only option and awaiting official graduation confirmation (semester end).
- GRE (entrance): 159 verbal, 149 quantitative, 4.0 analytical.
- 41 GPA credit hours, 3.51/4.0 GPA. 104 PhD Thesis Research credit hours.

B.S. in Computer Science, State University of New York at Plattsburgh *2015-2017*

- Dean's List honors in all five semesters.
- 17th Annual Computer Science Department Academic Excellence Award recipient (2016).
- 97 GPA credit hours, 3.86/4.0 GPA.

No degree, Utah Valley University, Germanna Community College, Northeastern University *1998, 2001-2002*

- Architecture, Arts and Sciences, & Engineering Focuses.
- 80 GPA credit hours.

Technical Skills

Languages: Python, JavaScript, PHP, HTML, CSS, Java, R, C, C++, C#, Swift, Assembly, VBA, VB.NET

Databases & Knowledge Graphs: MySQL, MariaDB, PostgreSQL, SQL Server, SQLite, Neo4j, Cypher, SPARQL, RDF, OWL, Protégé, WebProtégé, Virtuoso, RDFLib, pgAdmin, PhpMyAdmin, MySQL Workbench, ROBOT

Machine Learning & Data Science: PyTorch, TensorFlow, scikit-learn, Pandas, GeoPandas, NumPy, PySpark, Dask, Jupyter, Google Colab, matplotlib, domain adaptation, transfer learning, ADAPT, EasyAdapt, Balanced Weighting, CORAL, Subspace Alignment

Web Frameworks & Backend: Django, Flask, FastAPI, SQLAlchemy, Node.js, Next.js, serverless, REST, AJAX, Apache Cordova

Cloud & DevOps: AWS (RDS, S3, Lambda, ECS, EKS, CloudWatch, Route 53, IAM, Secrets Manager, Cognito), Azure (SQL, Key Vault, Fabric, Data Lake), Docker, Docker Compose, Podman, OpenStack, MassOpenCloud, CloudLab, n8n, Apache Web Server, Nginx, VPS

Web Scraping & Automation: Scrapy, cURL, HTTP/HTTPS, Browser Extensions, Greasemonkey, Userscripts, Chrome, n8n, tesseract

Visualization: D3.js, Chart.js, Three.js, LaTeX, ECharts, matplotlib, PowerBI, Tableau, Leaflet

Frontend: JavaScript, TypeScript, HTML/CSS, Bootstrap, React, templating (Django/Jinja), custom JavaScript libraries

Dev Tools & Platforms: Git, Bash/Shell, Linux (Fedora, Ubuntu, Debian), Android, GitHub Copilot, Chrome Extension API, SSH

HW/SW: Built and maintained PCs and laptops including Linux/Windows installation and maintenance.

Research Focus

Construction and use of knowledge graphs and ontologies, exploration of knowledge graph and ontology learning, and knowledge graph embeddings. Applying machine learning techniques in two key directions: 1) detecting bacteria species and small molecules in MALDI-TOF spectra, and 2) predicting concentrations of fecal indicator bacteria in coastal waters with transfer learning and domain adaptation.

Research Experience

Project Title: AI methods for reducing diver exposure to biological hazards
Research Advisors: Dr. Isabel Cruz, Dr. Cornelia Caragea
Duration: August 2021 - May 2026
Grant: ONR, #00014-21-1-2286

Project Title: A new paradigm for the creation and mining of microbial libraries for drug discovery
Research Advisor: Dr. Isabel Cruz
Duration: August 2018 - August 2021
Grant: NIH, #R01GM125943

Research Interests

Data science, web and data semantics, knowledge graphs, ontologies, embeddings, big data, machine learning, domain adaptation, transfer learning, databases, data mining, information retrieval, web search, information visualization, computational biology, open-source, agile development.

Teaching History

University of Illinois at Chicago - Chicago, IL — 2018-2024

Instructor
Data and Web Semantics, CS586 (2021)

Teaching Assistant
Digital Literacy, CS100 (2019)
Languages and Automata, CS301 (2019, 2023, 2024)
Software Design, CS342 (2018)
Software Engineering I, CS440 (2019)
Software Development for Mobile Platforms, CS478 (2020, 2021)
Big Data Mining, CS494 (2020)
Data and Web Semantics, CS586 (2019, 2020, 2021)

Graduate Coursework – UIC (relevant)

Computer Algorithms, CS401
Introduction to Machine Learning, CS412
Visualization and Visual Analytics I, CS424
Systems Performance and Concurrent Computing, CS463
Numerical Analysis, MCS471
Software Development for Mobile Platforms, CS478
Computer Algorithms II, CS501
Data and Web Semantics, CS586
Research Methods, CS590
Algorithms for Big Data, CS594
Visual Data Science, CS594

<u>Undergraduate Coursework – SUNY Plattsburgh (relevant)</u>
Artificial Intelligence, CSC345
Operating Systems, CSC433
Data Mining, CSC442
Software Design Studio, CSC446
Reliable Systems, CSC456
Climate Change Science, ENV406
Sustainable Transportation, GEG380
Urban Geography and Planning, GEG307
Linear Algebra I, MAT202
Sets, Functions and Relations, MAT231

<u>Manuscripts & Publications</u>
† co-author
* primary author
‡ supporting role

<u>Published</u>
* Shumway, D. (2026, July). Engineering a Quota-Constrained Human-in-the-Loop Data Platform: Marketplace-Aware Architecture over 3M+ Microtasks. Available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7040318
* Shumway, D. (2026, March). MALDI-DB: Architecture and Implementation of a Community-Driven Platform for Bacterial Identification via MALDI-TOF Mass Spectrometry. Technical report, University of Illinois at Chicago. Published online: https://davidshumway.github.io/self/MALDI-DB
† Elahi, A., Shumway, D., Kowalcyk, M., Shrestha, A., Caragea, D., Caragea, C., & Dorevitch, S. (2025). Machine Learning, Generalization, and Transfer Learning for Predicting the Exceedance of Fecal Indicator Bacteria Thresholds at Beaches. *Environmental Science & Technology, 59*(42), 22386-22396.
† Elahi, A., Shumway, D., Kowalcyk, M., Shrestha, A., Gautam, N., Caragea, D., Caragea, C., & Dorevitch, S. (2024, December). Predicting Surface Water Bacteria Levels Using Transfer Learning and Domain Adaptation. In *2024 IEEE International Conference on Big Data (BigData)* (pp. 1-10). IEEE.
† Gautam, N., Shumway, D., Kowalcyk, M., Khanal, S., Caragea, D., Caragea, C., & Dorevitch, S. (2023, April). Leveraging existing literature on the web and deep neural models to build a knowledge graph focused on water quality and health risks. In *Proceedings of the ACM Web Conference 2023* (pp. 4161-4171).

<u>Submitted & Revised</u>
† Submitted to *IJCAI (2023), DSAA (2023), The Web Conference (2024)*.
* Shumway, D., Elahi, A., Gautam, N., Kowalcyk, M., Caragea, D., Caragea, C., & Dorevitch, S. (2023). A Domain Adaptation Approach for Predicting Recreational Water Quality at Data-Scarce Coastal Locations. Submitted to *Environmental Science & Technology*.

<u>No Submission</u>
* Shumway, D., Elahi, A., Gautam, N., Kowalcyk, M., Caragea, D., Caragea, C., & Dorevitch, S. (2023). Developing a recreational and occupational waterborne illness knowledge graph toward biological risk assessment in coastal waters.
* Shumway, D. (2023). Use of large-scale public datasets toward improving coastal water quality predictions.
* Aggarwal, M., Shumway, D., Cruz, I. (2020). Bacteria Identification in MALDI-TOF Mass Spectrometry Data Using a Convolutional Neural Network Approach.
* Shumway, D. (2020). Storage and Querying of Provenance in Scientific Workflow Management Systems. Completed as part of PhD qualifying exam (UIC).
† Zhao, W., Shumway, D. (2020) A Review of QA4IE. Completed as part of Research Methods graduate course (UIC, CS590).
* Shumway, D. (2020). A public community-driven database for sharing and execution of matrix-assisted laser desorption ionization time-of-flight mass spectrometry bacterial and protein spectral data and workflows. Completed as part of Research Methods (UIC, CS590).
* Shumway, D. (2019). Bacterial identification in a distributed mass spectrometry sensor network using streaming big data. Completed as part of Algorithms for Big Data (UIC, CS594).
† Zhao, K., Shumway, D., Xingbo, W., Cruz, I. (2019). Efficient Multi-user Semantic Desktop Framework that Supports Data-aware Storage.
* Shumway, D. (2018). ADEPT - An ontology for Atomic Distributed Experiments: Modeling Scientific

Workflow Processes as Always-Available Cloud Services to aid in Workflow Reproducibility, Composition, Sharing, Resource Conservation, and Testability. Completed as part of Data and Web Semantics (UIC, CS586).

Grant Writing

‡ Dorevitch, S., Cruz, I., Catlett, C. (2021). Microbial hazard risk estimation and communication for Navy divers.

‡ Cruz, I., Miranda, F., Derrible, S., Siciliano, M., Cailas, M., Sambanis, A., … & McHenry, K. (2021). ICE–PRESUR: Illinois Center of Excellence–Planning a Resilient and Equitable State Using Real-time data.

† Cruz, I., Shumway, D. (2021). EarthCube Capabilities: Semantic Data Connections to Enable Distributed Science and Capability Building.

† Zhao, K., Shumway, D., Xingbo, W., Cruz, I. (2019). Efficient Knowledge Graph Processing in Modern Memory and Storage Hierarchy.

* Shumway, D. (2019). A collaboration and visualization tool to help meet sustainability goals by individuals and groups within local and global contexts. Completed as part of Grant Writing (UIC, UPP493).

Presentations

A public community-driven database for sharing and execution of matrix-assisted laser desorption ionization time-of-flight mass spectrometry protein fingerprints and workflows. Spring 2020. Presentation for PhD Written Critique and Proposal (WCP) PhD Qualifying Exam, UIC. Spring 2020.

A public community-driven database for sharing and execution of matrix-assisted laser desorption ionization time-of-flight mass spectrometry protein fingerprints and workflows. Spring 2020. Final presentation for Research Methods, UIC.

Bacterial identification in a distributed mass spectrometry sensor network using streaming big data. Spring 2019. Final presentation for Algorithms for Big Data, UIC.

ADEPT Ontology: Atomic Scientific Workflow Processes as Always-Available Cloud Services. Fall 2018. Presentation for Data and Web Semantics, UIC.

Natural Drug Discovery Web-Based Data Visualization: Project Proposal. Fall 2018. Presentation for Visual Data Science, UIC.

The Salvation of Paris: Notes on Sustainable Transit in Paris from Taras Grescoe's Straphanger. December 2017. Final presentation for Sustainable Transportation, SUNY Plattsburgh.

Containers, Docker, Linux: Notes from the 2016 LinuxCon Conference. August 2016. Presented to the Software Engineering Club at SUNY Plattsburgh.

Health and Well-Being in the Built Environment: Notes on Christopher Alexander. May 2016. Final presentation for Public Speaking, SUNY Plattsburgh.

Open Source Contributions

- Published Tools for Amazon's Mechanical Turk (Chrome, Firefox) on GitHub, a public productivity browser extension for AMT. Included documentation, issue tracking, and bugfixes. Used by up to 10,000 users per week.
- Publicly shared numerous small projects and code snippets on GitHub and GitHub Gists demonstrating a commitment to community and collaboration in software development.
- Contributed to open-source issue tracking through new issue documentation and creation of pull requests.

Conference Attendance

International Conference of Biomedical Ontologies, Virtual attendee. (2022).
EarthCube Annual Meeting, Virtual attendee. (2021, 2022).
UIC Urban Forum. (2019).
LinuxCon, Toronto. (2016).

Notebook Reviewer

2nd and 3rd Annual EarthCube Call for Notebooks. (2021, 2022).

Portfolio & Example Projects

LLM-Assisted Crowdsourcing Marketplace Codebase Analysis | Independent Research Project, 2026

- Developed an LLM-based microtask classification pipeline to categorize 500+ JavaScript automation

blocks across a 14-class domain taxonomy, engineering structured prompts with sliding-window conversation history and a domain-specific architectural cheat sheet to guide a local Llama3 model toward reliable JSON output.
- Built a git history redaction scanner to prepare an 8-year, 3,700-commit codebase for academic publication, combining regex-based detection of PII, API keys, and infrastructure fingerprints across all commit diffs with incremental save/resume, observability, folder-level filtering, and automated generation of a git filter-repo replacements file.

Tech: Claude, Llama 3B (local), GPU inference, Python, Jupyter, Git, Bash, JSON, embeddings, prompt engineering

Modular Insurance CRM Platform | Scalesology, 2025-2026
- Led backend development of a greenfield insurance CRM, migrating legacy workflows to a scalable Python/FastAPI architecture on AWS (250K+ LOC), including completion of roughly 8 out of 12 of the major backend system modules and design and implementation of 50-100 API endpoints
- Designed relational schema and managed PostgreSQL RDS infrastructure on AWS (RDS, EC2, S3); built core service modules supporting complex CRM workflows via SQLAlchemy and n8n.
- Collaborated cross-functionally to translate evolving client requirements into backend APIs and data models; proactively flagged and mitigated system-level design risks.
- Leveraged LLMs throughout the development lifecycle, generating PRs (GitHub Copilot), system design, documentation, unit tests, and FastAPI/SQLAlchemy code, and iterated on Copilot PR feedback until optimal solutions were achieved within time constraints, with targeted human tweaks to ensure full integration.
- Maintained a sustained contribution cadence of 150-250 commits per month over 7 months, reflecting deep ownership of the codebase and continuous delivery of features.

Tech: Python, FastAPI, SQLAlchemy, PostgreSQL, AWS (RDS, ECS, EKS, S3, CloudWatch, Route 53, IAM, Secrets Manager, Cognito), n8n, Git, GitHub Copilot, React

React Joyride Tour + Components for University Recruiting App | Scalesology, 2024-2025
- Added React Joyride tour functionality to a custom university recruiting app (roughly 10 guided tour steps), working alongside a lead React developer and backend engineer (SQLAlchemy, PostgreSQL on RDS).
- Held sole responsibility for tour development, collaborating with PM/SA to design and implement client-desired functionality.
- Used D3.js alongside ECharts to implement a custom ECharts parliament chart relating the ratio of students retained versus left.

Tech: React, React Joyride, JavaScript, SQLAlchemy, PostgreSQL, AWS RDS, ECharts, D3.js

YAML Unit Testing Framework Extension | Scalesology, 2024-2025
- Added new YAML unit tests and modified the custom Python unit test runner to enable execution against additional database schemas (prod, sandbox, dev, etc.) from the command line.
- Ensured consistent formatting and readability across existing and new YAML tests for long-term maintainability.

Tech: Python, YAML, PostgreSQL

Oil Recycling Work Order Processing Migration | Scalesology, 2024-2025
- Transitioned ~50% of legacy SQL Server stored procedures to Python for work order processing, adding readability, maintainability, and structured logging to the system.
- Maintained SQL Server-to-Snowflake migration scripts and diagnosed warnings and errors.
- Took over PowerBI administration and reporting after client's analyst departed, performing both administrative duties such as enabling access for new users, and new PowerBI development such as addressing bugfixes and adding new report functionality.
- Analyzed Snowflake data warehouse usage and reviewed DAX queries for Oracle migration planning.
- Diagnosed Snowflake security and access issues, including differentiating human vs. programmatic accounts, enforcing two-factor authentication for human users by generating and sending 2FA enrollment emails, and performing a full accounting of all human accounts on Snowflake while ensuring all Python Snowflake libraries were up-to-date.

Tech: Python, SQL Server, Snowflake, PowerBI, DAX, SQL

Tableau Reports for Financial Company | Scalesology, 2025

- Created and evolved Tableau reports tracking financial performance data for companies across two separate reports, with multiple visualizations, pages, and tabs per report.
- Built a Python ETL script to process messy spreadsheet data (copied from an external reporting tool) into clean Tableau-ready format, transforming block-style report data into standardized rows and columns.
- Data was loaded via Excel spreadsheets with multiple tabs containing company performance metrics.

Tech: Tableau, Excel, Python

Grocery Store PowerBI Proof of Concept | Scalesology, 2025

- Transitioned a large JavaScript + SQL Server report into a PowerBI proof of concept with multiple pages, tables, charts, and a custom data matrix visualization (compiled via terminal).
- Loaded SQL Server data (30-60K grocery products) into Azure SQL, then pulled into PowerBI and Fabric/OneLake for high-speed queries across millions of rows with semantic modeling and performance tuning.

Tech: PowerBI, Fabric, OneLake, Azure SQL, SQL Server, custom visualizations

Building Footprint Clustering Tool (DBSCAN → HDBSCAN) | Scalesology, 2025

- Evolved a building footprint clustering tool from DBSCAN to HDBSCAN to add confidence scores to algorithm outputs for an insurance client.
- Added Microsoft building dataset to the app using PostGIS on AWS RDS for an additional building footprint heuristic.
- Discovered and fixed redundant geocoding API calls by adding disk-based caching keyed by request parameters, reducing API costs and accelerating development iteration.

Tech: Python, HDBSCAN, PostGIS, AWS RDS, geocoding APIs

AI methods for reducing diver exposure to biological hazards | Lead Developer, 2021-5/2026

- Constructed water quality datasets by combining multiple environmental data, developed and tested machine learning, transfer learning, and domain adaptation models, including supervised, semi-supervised, and unsupervised methods, and helped to build a waterborne illness ontology.

Tech: Python, LaTeX, SPARQL, RDF, OWL, Protégé, WebProtégé, Virtuoso, Dask, rdflib, OBO ROBOT, Overleaf

Visualization & Data Analytics: Pandas, matplotlib, GeoPandas, D3.js, Plotly, Google Colab + Google Drive, Google Sheets

Machine Learning & AI: time-series prediction, domain adaptation, transfer learning, supervised & unsupervised learning, PyTorch, TensorFlow, scikit-learn, Linear Regression, Random Forest, Decision Tree, SVM, MLP, CNN, XGBoost, LSTM, ADAPT, EasyAdapt, subspace alignment, CORAL, GRU, GNN, BERT, LLM

Popular APIs & Datasets: UMLS, Infectious Disease Ontology, WaterQualityData.us, EPA BEACON, OpenStreetMap API, NOAA Global Hourly, NOAA Tides & Currents, NOAA NDBC, EPA ECHO, EN4 Met Office Hadley Centre Ocean Salinity

FDIC Quarterly Banking App | Independent project, 2023

- Single-page app using Flask, JavaScript, and DataTables.js to analyze FDIC Banks API reporting data from 5K+ US financial institutions, with mechanisms for automated retrieval, caching, and CSV download of selected views.

Tech: Flask, DataTables.js, FDIC Banks API

MALDI-DB: A public repository for MALDI-TOF mass spectra | Lead Developer, 2018-2021

- Built a community-driven public repository for preprocessing, analysis, search, and hosting of MALDI-TOF mass spectra and their small molecules, utilizing R, RPlumber, PostgreSQL, Nginx, Python, Django, D3.js, Docker, Docker Compose.
- Utilized OpenStack on MassOpenCloud to provision resources and publish a development version of the site online.

Files: https://github.com/idbac/maldidb
Development screenshots: https://github.com/idbac/maldidb/tree/master/mdb/media

Zoomie: Round-robin breakout scheduling in Zoom | Independent project, 2020

- Simple browser extension built for a meditation group which utilizes Zoom's web-chat client, enabling moderators to run round-robin partnered breakout room events (Firefox, Chrome), open sourced on GitHub with documentation, issue tracking, and bugfixes.

Chicago Public Schools Enrollment App | Visual Design Studio (CS526), 2018
- Single-page app built the UIC Department of Minority Affairs toward recruitment of high school students from within the City of Chicago.
- Led the development of 3 interactive visualizations and search functionality, then took the initiative to refactor the entire project into a seamless single-page application over winter break, resulting in a grade upgrade and the app being adopted as a teaching demo.

Tech: HTML, CSS, JavaScript, D3.js, Leaflet, Chicago Public Schools dataset
Files: https://github.com/davidshumway/CPS

Finger Lakes Business Systems | Lead Developer, 2010-2018
Tech: JavaScript, Linux, MySQL, PHP, HTML, CSS, Python, Google Chrome API, Chrome/Firefox browser extensions, Amazon Mechanical Turk (AMT), Chart.js
Files (available on request):
- AMT marketplace-related browser extensions
- AMT marketplace-related MySQL databases
- AMT marketplace-related PHP scripting
- AMT marketplace-related Apache Cordova mobile app
- AMT task-related browser extensions
- AMT task-related MySQL databases
- AMT task-related PHP scripting
- AMT task-related Python scripting (Scrapy)
- Browser extensions, SQL, and PHP scripting related to clients outside of AMT

Popular APIs used included Google Places (nearby search, text search, autocomplete, radar search), Google Maps, Yelp Businesses (v1-v3), Facebook Graph, DBPedia, Foursquare, BBB, Wikipedia, GeoNames, Twitter, Airbnb, Upcdatabase.org, Smoopa.com.

Tools for Amazon's Mechanical Turk | Independent project, 2013-2018
- Developed public-facing open-source Firefox and Chrome browser extensions to improve worker productivity on the (AMT) website, utilized by up to 10,000 users per week.

Tech: Firefox / Chrome browser extension, Chrome Extension API, HTML, CSS, JavaScript

Trumpocalypse | Team project, Software Design Studio (CSC446), Fall 2017
- Politically-themed satirical text-based game in the style of Oregon Trail, built using the Python Pygame library.
- Completed roughly 350 out of 500 team commits.

École Polytechnique Fédérale de Lausanne | Summer Intern, Summer 2017
Tech: Neo4j, Cypher, Ubuntu, Node.js, PostgreSQL
Files (available upon request):
- Cypher queries for working with Kamusi’s Neo4j graph database
- PostgreSQL queries for analysis of the PanLex language database

Paul Brunton Philosophic Foundation | Technology Specialist & Developer, 2010-2013
- Automated the (EPSON) scanning process, resulting in 4-8x efficiency and saving 2-4 years of manual scanning work.
- Created MySQL database full-text search form within Microsoft Word using VB.NET and Visual Studio.
- Utilized tesseract open-source OCR library to extract text from typewriter-written documents, saving the extracted text to a MySQL database, and implemented tools for 1) LibreOffice auto-complete using macros, and 2) duplicate hard-copy document checking.
- Utilized ImageMagick to automatically crop scanned white space from thousands of scanned typewritten documents, and created and used an interactive tool to validate cropped white space using thumbnails (HTML/JavaScript/CSS), providing performance gains of up to 5x over manual image cropping.

Tech: Linux, Apache, MySQL, PHP, HTML, CSS, JavaScript, VBA, VB.NET, Tesseract, Bash, Visual Studio, LibreOffice, Microsoft Word
Files (available on request):
- Epson scanning automation tool: Initially VBA macros, and later combined with VB.NET front-end
- Microsoft Word, MySQL database full-text search form (VB.NET)
- Tesseract OCR via shell scripting, MySQL database
- ImageMagick cropping and crop-review tool (LAMP, HTML, JavaScript, CSS)

Freelance Technology Specialist | Technology Specialist & Developer. 2005-2010
- Provided freelance tech services in person and on freelance sites such as Rent-A-Coder, Freelancer.com, and MTurk.com.

Tech: Linux, Apache, MySQL, PHP, HTML, CSS, JavaScript, Audio transcription